

MIST: A Multi-Context Spatial Transformer for Gene Expression Prediction from Histology

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Abstract

Spatial transcriptomics reveals how gene expression is organized across a tissue; however, it is expensive and time-consuming. This motivates predicting gene expression from standard H&E images. A spot’s morphology, its spatial neighborhood, and the broader tissue context provide complementary information for predicting gene expression from H&E images. A key modeling challenge is how to integrate these cues effectively. We introduce MIST, a compact spatial transformer that predicts a slide’s spatial gene expression in a single forward pass by fusing three complementary views of context in every layer: local attention over spatial nearest neighbors, global self-attention across all spots, and an attention-pooled slide representation. Because the model uses spot coordinates only through pairwise distances, its predictions are invariant to translations, rotations, and reflections of the coordinate frame when image features are fixed. On six human organ cohorts from HEST-1k, evaluated under intra-cohort, pooled multi-organ, and leave-one-organ-out settings, MIST attains the highest mean Pearson correlation (PCC) on every cohort, improving over the strongest baseline by 9.7%, 11.6% and 11.4% for the 10-, 50-, and 100-gene panels, respectively, with the lowest MSE and MAE. Moreover, the gains are statistically significant and persist across the evaluated gene-selection strategies (HVG, DEG, and HMMVG) and encoder features (UNI, CONCH, and UNI+CONCH). It generalizes to held-out organs (+6.0% over the closest baseline BLEEP). MIST also better preserves tissue spatial structure: when we group spots into regions (spatial domains) from the predicted expression, these regions match those obtained from the measured expression more closely than for BLEEP, on 78% of the 58 held-out slides by the Adjusted Rand Index (ARI) and 90% by Normalized Mutual Information (NMI), two standard measures of agreement between clusterings. It also reproduces spatial expression patterns of selected melanoma-associated genes (mean PCC 0.90) and breast cancer receptor genes associated with HER2, ER, and PR, supporting its ability to capture biologically relevant molecular patterns from H&E images.

The exact coordinate guarantee assumes consistent resolution of k NN distance ties. On held-out CCRCC and PRAD slides, rigid coordinate transforms preserve aggregate PCC to within 0.000166, although individual predictions can change.

1 Introduction

Spatial transcriptomics (ST) measures gene expression together with tissue location, enabling the study of spatial heterogeneity, tissue organization, and cellular interactions [Ståhl et al. \(2016\)](#). Its cost and experimental burden, however, constrain the number and diversity of specimens that can be profiled. In contrast, H&E slides are routinely collected and contain morphological signals associated with tissue composition and molecular state [Chen et al. \(2024a\)](#); [Jaume et al. \(2024\)](#). Predicting spatial gene expression from these images

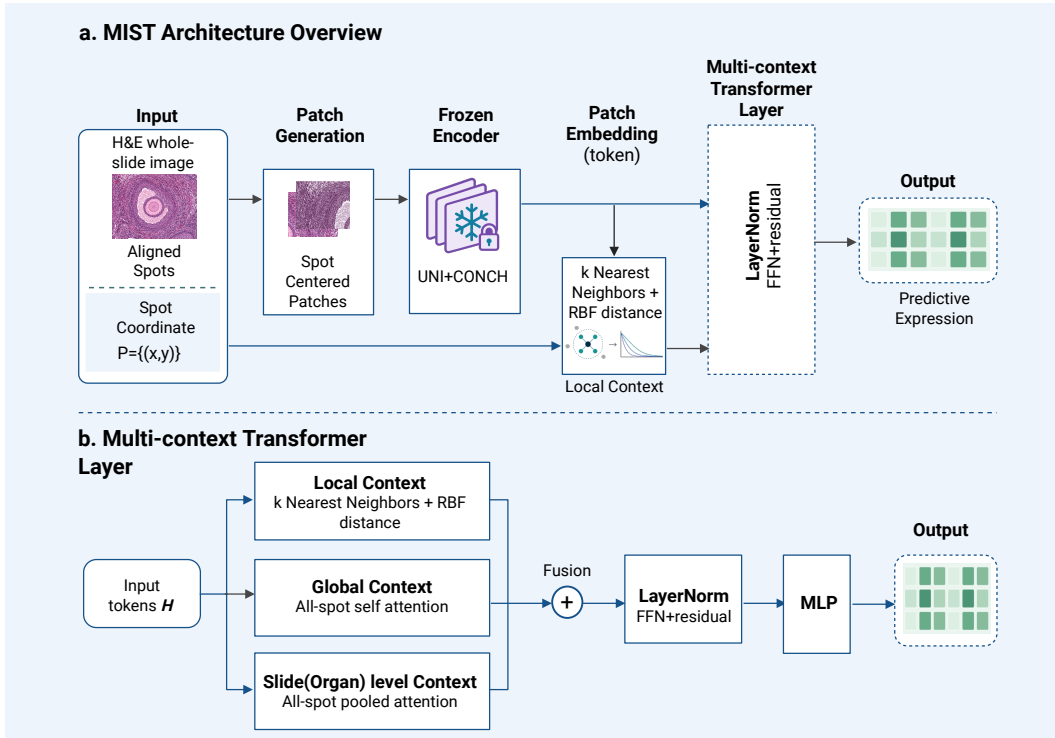


Figure 1: MIST overview. (a) Frozen UNI, CONCH, or combined features turn H&E spot patches into tokens, while coordinate-based k NN edges provide local context. (b) Each layer combines distance-biased local attention with global self-attention before the expression head. Pairwise-distance coordinates ensure invariance to translations, rotations, and reflections.

could therefore extend molecular analysis to specimens without matched ST measurements He et al. (2020); Zeng et al. (2022); Xie et al. (2023).

Morphology and expression interact locally, across the slide, and at the level of slide composition. Existing predictors capture parts of this structure with patch regression, spatial graphs, retrieval, or conditional generation He et al. (2020); Zeng et al. (2022); Xie et al. (2023); Zhu et al. (2025), but absolute coordinates can make predictions depend on an arbitrary scanner frame. Geometric representations based on relative quantities provide a principled way to reduce this dependence Satorras et al. (2021). We therefore seek a compact predictor that combines multi-scale context while using coordinates only through relative distances.

We introduce MIST, a direct spatial transformer that represents each spot as a token and combines spatial k NN attention for local context, full self-attention for slide-wide context, and attention pooling for a slide summary Vaswani et al. (2017). Because spot coordinates enter the model only through pairwise distances, its predictions are invariant to translations, rotations, and reflections when image features are fixed and k NN ties are resolved consistently (Fig. 1).

Main idea. MIST predicts spatial gene expression by combining three complementary messages at every spot: distance-aware self-attention over nearby spots, standard global self-attention over all spots, and an attention-pooled summary of the overall tissue composition.

We evaluate MIST across six human cohorts under intra-cohort prediction, pooled multi-organ prediction, and leave-one-organ-out transfer. In intra-cohort evaluation, MIST achieves the highest mean Pearson correlation and the lowest mean squared error (MSE) and mean absolute error (MAE), averaged across cohorts, for all evaluated gene-selection strategies and panel sizes. It also improves transfer to held-out organs and preserves biologically meaningful spatial patterns for melanoma- and breast-cancer-associated genes. Finally, on

58 held-out slides, MIST improves spatial-domain recovery over BLEEP on 78% of slides according to ARI and on 90% according to NMI, although absolute domain agreement remains modest.

Our experiments address five research questions: RQ1 (Performance): How accurately does MIST predict gene expression across organs when trained separately per cohort or jointly across cohorts? RQ2 (Transfer): How well does it generalize to an organ excluded from training? RQ3 (Ablation): Which context pathways and spatial-neighborhood choices drive predictive accuracy? RQ4 (Coordinate invariance): How stable are predictions and accuracy under transformations of the spot-coordinate frame? RQ5 (Structure): How well do predicted expression profiles preserve spatial domains identified from measured expression, as measured by ARI and NMI?

To sum up, we have the following summarized contributions:

1. We introduce MIST, a compact multi-context spatial transformer that integrates local neighborhood attention, global slide-wide attention, and slide-level context using continuous pairwise-distance features and coordinate-frame invariance.
2. We establish a controlled evaluation protocol spanning intra-cohort, pooled cross-organ, and held-out-organ settings, with matched gene panels, image features, splits, baselines, and model selection.
3. We show that MIST achieves strong predictive performance across organs, gene-selection strategies, panel sizes, and encoder features, while remaining a deterministic single-pass predictor.
4. We evaluate downstream biological structure, showing improved spatial-domain recovery according to ARI and NMI and accurate reconstruction of cancer-associated marker-gene patterns.

2 Related Work

H&E-to-ST prediction. Existing methods include patch-level regression (ST-Net [He et al. \(2020\)](#)), spatial graph models (HisToGene [Pang et al. \(2021\)](#); Hist2ST [Zeng et al. \(2022\)](#)), cross-modal retrieval (BLEEP [Xie et al. \(2023\)](#)), and multi-resolution or multi-cell approaches (TRIPLEX [Chung et al. \(2024\)](#); M2OST [Wang et al. \(2025\)](#)). STEM [Zhu et al. \(2025\)](#) instead models expression conditionally with diffusion. Because published results use different features, panels, and splits, we compare these model families under a matched protocol; MIST remains a deterministic point predictor rather than a distributional model.

Context modeling. Spatial predictors differ in how they aggregate context: some use only local patches, whereas others add neighboring-spot graphs, nonlocal image information, or multi-resolution features. MIST combines distance-aware attention over local neighbors, standard self-attention over all spots, and an attention-pooled tissue summary in every layer.

Pathology foundation models. Recent pathology encoders provide transferable representations for morphology and tissue composition [Chen et al. \(2024b\)](#); [Lu et al. \(2024\)](#). We use frozen UNI and CONCH features as matched inputs for MIST and the baselines, and separately evaluate encoder choice to distinguish representation quality from architectural effects.

Geometry and generalization. Absolute position encodings can change when an equivalent tissue layout is translated, rotated, or reflected; geometric networks instead construct interactions from relative quantities [Satorras et al. \(2021\)](#). MIST uses pairwise Euclidean distances, which are invariant to these frame changes for fixed histology features. We also separate pooled cross-validation, where organs remain represented in training, from leave-one-organ-out (LOOO) evaluation on HEST-1k [Jaume et al. \(2024\)](#) to measure anatomical transfer. Further positioning details appear in Appendix [D](#).

3 Method

MIST maps frozen spot-level histology features and measured spot coordinates to a spatial gene expression map. Its method has three stages: coordinate-invariant distance encoding, a shared multi-context layer, and supervised prediction with a correlation-aware objective.

3.1 Problem Setup

For a slide with N spots, let $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_N]^\top \in \mathbb{R}^{N \times d_{\text{in}}}$ denote frozen pathology features and $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_N]^\top \in \mathbb{R}^{N \times 2}$ the corresponding spot coordinates. The target $\mathbf{Y} \in \mathbb{R}^{N \times G}$ contains log-normalized expression for G genes. We learn $f_\theta : (\mathbf{F}, \mathbf{P}) \mapsto \hat{\mathbf{Y}}$ and process each slide independently; the model therefore supports a variable number of spots and does not require expression measurements at test time. In the feature-matched main comparisons, $\mathbf{f}_i$ concatenates frozen UNI [Chen et al. \(2024b\)](#) and CONCH [Lu et al. \(2024\)](#) embeddings ($d_{\text{in}} = 1536$).

3.2 Coordinate-Invariant Distance Encoding

We first compute the pairwise Euclidean distance matrix $D_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$. For each spot i , we retain the indices $\mathcal{N}(i)$ and distances of its k nearest neighbors, excluding the spot itself. Distances are expanded into M radial-basis features,

$$\phi(D_{ij}) = [\exp(-\gamma_m D_{ij}^2)]_{m=1}^M, \quad (1)$$

where the fixed bandwidth parameters γ_m are logarithmically spaced. These features influence attention weights but are never added to the spot embeddings as absolute positions.

Proposition 1 (Coordinate frame invariance):

For fixed model parameters and fixed histology features, in deterministic evaluation mode, let $R \in \mathbb{R}^{2 \times 2}$ be orthogonal, let $\mathbf{t} \in \mathbb{R}^2$, and define $\mathbf{P}' = \mathbf{P}R^\top + \mathbf{1}\mathbf{t}^\top$. If the k NN construction resolves distance ties consistently, the prediction is unchanged:

$$f_\theta(\mathbf{F}, \mathbf{P}') = f_\theta(\mathbf{F}, \mathbf{P}). \quad (2)$$

Details, qualifications, and the proof of this proposition are provided in [Supplementary Section B](#).

3.3 Multi-Context MIST Layer

Core Idea. At every layer, each spot receives three messages in parallel: one from its nearby spots, one from all spots, and one from a pooled summary of the slide. The model adds these messages to the current spot representation and refines the result with a feed-forward network. Only the nearby-spot message uses coordinates, and it uses pairwise distances rather than absolute positions.

The input features are projected to hidden width d , $\mathbf{X}^{(0)} = \mathbf{F}W_{\text{in}} + \mathbf{b}_{\text{in}}$. The same neighbor graph and radial-basis features are reused in all L transformer layers. At layer ℓ , a pre-normalized representation $\mathbf{H} = \text{LN}(\mathbf{X}^{(\ell)})$ is passed through three context pathways.

3.3.1 Local spatial-neighborhood attention

Each spot attends to its k neighbors with a learned, distance-dependent bias for each attention head. For head h , the attention weights are

$$\alpha_{ij}^h = \text{softmax}_{j \in \mathcal{N}(i)} \left(\frac{(\mathbf{q}_i^h)^\top \mathbf{k}_j^h}{\sqrt{d_h}} + b_h(\phi(D_{ij})) \right), \quad (3)$$

where $d_h = d/H$ for H heads. The local output is the concatenation of the weighted value sums, followed by a learned output projection. Morphological similarity is captured by the

query-key term, while spatial distance can contribute through the learned radial-basis bias. In the reported native-coordinate HEST configuration, neighbor distances are large enough that the RBF values are negligible, so the main practical contribution of this pathway is attention over the spatial k NN neighborhood; the continuous-distance bias is evaluated in the corresponding ablations.

3.3.2 Global self-attention

In parallel, standard multi-head self-attention [Vaswani et al. \(2017\)](#) allows every spot to attend to all N spots. This pathway captures long-range dependencies that may not be connected in the local spatial graph. It introduces no direct coordinate-dependent bias and is permutation-equivariant with respect to spot order; in later layers, its inputs may nevertheless carry spatial information propagated by the local and slide-context pathways.

3.3.3 Attention-pooled slide context

A learned scalar scoring function produces $a_i = \text{softmax}_i(s(\mathbf{h}_i))$. The slide representation is

$$\mathbf{g} = W_g \left(\sum_{i=1}^N a_i \mathbf{h}_i \right) + \mathbf{b}_g, \quad (4)$$

and is broadcast to every spot. Unlike a learned class token, this representation is computed directly from the slide’s current spot representations and summarizes the slide-level context.

For the complete model, the local ($\mathbf{L}$), global ($\mathbf{G}$), and broadcast slide ($\mathbf{S}$) representations are summed and added to the residual stream. A second pre-normalization residual update applies a two-layer feed-forward network with expansion factor four and GELU activation:

$$\tilde{\mathbf{X}}^{(\ell)} = \mathbf{X}^{(\ell)} + \mathbf{L}^{(\ell)} + \mathbf{G}^{(\ell)} + \mathbf{S}^{(\ell)}, \quad (5)$$

$$\mathbf{X}^{(\ell+1)} = \tilde{\mathbf{X}}^{(\ell)} + \text{FFN}(\text{LN}(\tilde{\mathbf{X}}^{(\ell)})). \quad (6)$$

After L layers, layer normalization and a linear head map each spot representation to its G predicted gene values. The model produces the complete slide prediction in one forward pass.

Proposition 2 (Permutation equivariance):

Let $Q \in \{0, 1\}^{N \times N}$ be a permutation matrix. If the k NN construction handles distance ties equivariantly, then

$$f_\theta(Q\mathbf{F}, Q\mathbf{P}) = Qf_\theta(\mathbf{F}, \mathbf{P}). \quad (7)$$

Therefore, reordering the input spots only reorders their predicted expression profiles. [Supplementary Section B](#) provides the proof. Our numerical checks evaluate both Propositions 1 and 2 under finite precision.

3.4 Training Objective and Ablations

For a slide with measured expression $\mathbf{Y}$, we optimize mean squared error together with a gene-wise Pearson-correlation term:

$$\mathcal{L} = \text{MSE}(\hat{\mathbf{Y}}, \mathbf{Y}) + \lambda \left(1 - \frac{1}{|\mathcal{V}|} \sum_{g \in \mathcal{V}} \rho_g \right), \quad (8)$$

where ρ_g is the Pearson correlation between predicted and measured expression across the N spots, $\mathcal{V}$ is the set of genes with nonzero target variance on that slide, and $\lambda = 0.5$. MSE constrains the scale of the predictions, while the correlation term encourages the model to reproduce spot-to-spot expression patterns.

The complete model uses all three context pathways. Their individual contributions are evaluated in the factorial ablation study reported in Section 5.3, which compares local (L), global (G), and slide-level (S) context configurations.

4 Experimental Setup

Data and preprocessing. We evaluate on HEST-1k [Jaume et al. \(2024\)](#) using frozen UNI+CONCH spot features (1536 dimensions) and $\log(1+x)$ -transformed expression targets. The cross-organ analyses use six cohorts: CCRCC, IDC, LUNG, PAAD, PRAD, and SKCM; HCC, LYMPH_IDC, and READ are excluded. Additional cohort details are provided in Appendix C.

Gene-panel selection. Unless otherwise stated, the default target is the 50-gene HEST HVG panel, whose genes are selected by highest expression variance. To test sensitivity to gene-selection strategy, we also evaluate 50-gene DEG and HMHVG panels. DEG genes are selected using differential expression across Leiden clusters, whereas HMHVG combines mean-expression and variability criteria after variance filtering (Table 1). We additionally evaluate panel sizes of 10, 50, and 100 genes.

Evaluation regimes. We consider intra-cohort training, in which a separate model is trained for each cohort; pooled cross-organ training, in which slide-level samples are held out while organ types remain represented in training; and leave-one-organ- out (LOOO) transfer, in which the test organ is absent from training. Organ identity is not provided as a model input.

Metrics and validation. Our primary metric is mean per-gene Pearson correlation (PCC). We also report MSE and MAE for expression-scale accuracy, and ARI and NMI for spatial-domain recovery. Efficiency is measured using inference latency and peak GPU memory. Checkpoints are selected on inner validation folds and evaluated once on the outer test fold; results are averaged over three seeds.

Baselines and implementation. ST-Net [He et al. \(2020\)](#), Hist2ST [Zeng et al. \(2022\)](#), BLEEP [Xie et al. \(2023\)](#), and STEM [Zhu et al. \(2025\)](#) are evaluated under matched features, panels, splits, budgets, and model-selection rules. Unless otherwise stated, MIST uses four layers, width 256, eight spatial neighbors, and 4.88M trainable parameters. Full preprocessing, split, hyperparameter, and implementation details are provided in Appendix F.

5 Results

5.1 RQ1: Predictive performance across diverse organs

We begin by asking whether MIST predicts spatial expression accurately on organs it has seen during training. To answer this, we measure accuracy in two complementary ways under matched features, panels, and splits: a separate model trained per cohort (intra-cohort), and a single model trained jointly across all organs (pooled).

In the intra-cohort setting, MIST is best under every 50-gene selection strategy (Table 1), with mean PCCs of 0.496 (HVG), 0.615 (DEG), and 0.650 (HMHVG), compared with BLEEP’s 0.444, 0.559, and 0.587. The advantage also holds across panel sizes (Table 2): MIST reaches PCC@10/50/100 values of 0.600/0.496/0.559 and wins all six cohorts at PCC@50, yielding an 11.6% gain over BLEEP. It also has the lowest MSE and MAE under every tested strategy and panel size (Table 2).

The intra-cohort improvement is statistically reliable: across six cohorts and three seeds ($n=18$ paired runs), the PCC@50 gain over BLEEP is 0.051 (95% bootstrap CI [0.041, 0.062]; Wilcoxon $p=7.6 \times 10^{-6}$). At the per-gene level, MIST outperforms BLEEP on 87% of genes ($n=300$; Wilcoxon $p < 10^{-30}$). A summary of the paired statistical tests is reported in Appendix E.

Table 1: Intra-cohort HEST-1k: 50-gene selection strategies. Dataset PCC is mean $\pm$ SD over three seeds; final columns are six-cohort mean PCC, MSE, and MAE. Best per strategy in bold; $\uparrow$ higher and $\downarrow$ lower is better.

Panel	Model	IDC	CCRCC	LUNG	PAAD	PRAD	SKCM	PCC Avg. $\uparrow$	MSE Avg. $\downarrow$	MAE Avg. $\downarrow$
HVG	ST-Net	0.551 $\pm$ 0.01	0.163 $\pm$ 0.02	0.521 $\pm$ 0.01	0.366 $\pm$ 0.03	0.287 $\pm$ 0.02	0.661 $\pm$ 0.01	0.425	1.995	0.969
	Hist2ST	0.305 $\pm$ 0.11	0.180 $\pm$ 0.03	0.430 $\pm$ 0.05	0.248 $\pm$ 0.04	0.323 $\pm$ 0.01	0.523 $\pm$ 0.00	0.335	1.561	0.890
	BLEEP	0.519 $\pm$ 0.00	0.175 $\pm$ 0.02	0.565 $\pm$ 0.01	0.412 $\pm$ 0.03	0.346 $\pm$ 0.01	0.650 $\pm$ 0.00	0.444	1.318	0.819
	STEM	0.434 $\pm$ 0.00	0.118 $\pm$ 0.01	0.407 $\pm$ 0.00	0.233 $\pm$ 0.02	0.207 $\pm$ 0.02	0.461 $\pm$ 0.02	0.310	1.747	0.920
	MIST (Ours)	0.581$\pm$0.01	0.252$\pm$0.01	0.589$\pm$0.01	0.437$\pm$0.03	0.410$\pm$0.01	0.707$\pm$0.01	0.496	1.233	0.794
DEG	ST-Net	0.697 $\pm$ 0.01	0.312 $\pm$ 0.01	0.566 $\pm$ 0.01	0.425 $\pm$ 0.03	0.413 $\pm$ 0.00	0.752 $\pm$ 0.01	0.528	3.336	1.457
	Hist2ST	0.549 $\pm$ 0.07	0.335 $\pm$ 0.01	0.518 $\pm$ 0.01	0.352 $\pm$ 0.01	0.437 $\pm$ 0.01	0.656 $\pm$ 0.00	0.475	1.757	1.021
	BLEEP	0.731 $\pm$ 0.01	0.304 $\pm$ 0.00	0.621 $\pm$ 0.01	0.487 $\pm$ 0.03	0.466 $\pm$ 0.00	0.746 $\pm$ 0.03	0.559	1.557	0.945
	STEM	0.684 $\pm$ 0.02	0.253 $\pm$ 0.01	0.440 $\pm$ 0.01	0.306 $\pm$ 0.04	0.355 $\pm$ 0.03	0.617 $\pm$ 0.02	0.443	1.958	1.054
	MIST (Ours)	0.774$\pm$0.01	0.402$\pm$0.01	0.662$\pm$0.00	0.518$\pm$0.03	0.539$\pm$0.01	0.794$\pm$0.00	0.615	1.477	0.928
HMHVG	ST-Net	0.724 $\pm$ 0.01	0.339 $\pm$ 0.01	0.568 $\pm$ 0.00	0.441 $\pm$ 0.03	0.452 $\pm$ 0.00	0.824 $\pm$ 0.03	0.558	3.589	1.470
	Hist2ST	0.633 $\pm$ 0.08	0.387 $\pm$ 0.01	0.598 $\pm$ 0.00	0.341 $\pm$ 0.04	0.477 $\pm$ 0.01	0.710 $\pm$ 0.14	0.524	1.832	1.042
	BLEEP	0.739 $\pm$ 0.02	0.349 $\pm$ 0.01	0.661 $\pm$ 0.00	0.493 $\pm$ 0.02	0.432 $\pm$ 0.04	0.849 $\pm$ 0.02	0.587	1.566	0.936
	STEM	0.687 $\pm$ 0.01	0.290 $\pm$ 0.01	0.517 $\pm$ 0.01	0.305 $\pm$ 0.03	0.337 $\pm$ 0.02	0.731 $\pm$ 0.06	0.478	1.976	1.049
	MIST (Ours)	0.774$\pm$0.01	0.454$\pm$0.02	0.693$\pm$0.00	0.524$\pm$0.02	0.555$\pm$0.02	0.901$\pm$0.00	0.650	1.393	0.901

Table 2: Intra-cohort HEST-1k: gene-panel sizes. Dataset PCC is mean $\pm$ SD over three seeds; final columns are six-cohort mean PCC, MSE, and MAE. Best per panel size in bold; $\uparrow$ higher and $\downarrow$ lower is better.

Panel size	Model	IDC	CCRCC	LUNG	PAAD	PRAD	SKCM	PCC Avg. $\uparrow$	MSE Avg. $\downarrow$	MAE Avg. $\downarrow$
PCC@10	ST-Net	0.666 $\pm$ 0.01	0.231 $\pm$ 0.01	0.702 $\pm$ 0.01	0.511 $\pm$ 0.04	0.442 $\pm$ 0.03	0.728 $\pm$ 0.00	0.547	6.087	1.908
	Hist2ST	0.636 $\pm$ 0.01	0.273 $\pm$ 0.02	0.519 $\pm$ 0.02	0.377 $\pm$ 0.07	0.459 $\pm$ 0.02	0.471 $\pm$ 0.04	0.456	3.297	1.374
	BLEEP	0.635 $\pm$ 0.02	0.231 $\pm$ 0.02	0.735 $\pm$ 0.00	0.536$\pm$0.03	0.434 $\pm$ 0.03	0.697 $\pm$ 0.01	0.545	2.859	1.243
	STEM	0.591 $\pm$ 0.02	0.170 $\pm$ 0.00	0.616 $\pm$ 0.01	0.393 $\pm$ 0.04	0.325 $\pm$ 0.03	0.598 $\pm$ 0.01	0.449	3.181	1.310
	MIST (Ours)	0.708$\pm$0.00	0.322$\pm$0.01	0.759$\pm$0.00	0.519 $\pm$ 0.05	0.524$\pm$0.02	0.769$\pm$0.01	0.600	2.620	1.211
PCC@50	ST-Net	0.551 $\pm$ 0.01	0.163 $\pm$ 0.02	0.518 $\pm$ 0.01	0.366 $\pm$ 0.03	0.294 $\pm$ 0.02	0.665 $\pm$ 0.01	0.425	1.995	0.969
	Hist2ST	0.309 $\pm$ 0.07	0.184 $\pm$ 0.02	0.470 $\pm$ 0.01	0.248 $\pm$ 0.04	0.333 $\pm$ 0.02	0.522 $\pm$ 0.03	0.335	1.561	0.890
	BLEEP	0.519 $\pm$ 0.00	0.175 $\pm$ 0.02	0.562 $\pm$ 0.01	0.412 $\pm$ 0.03	0.348 $\pm$ 0.01	0.645 $\pm$ 0.01	0.444	1.318	0.819
	STEM	0.439 $\pm$ 0.01	0.123 $\pm$ 0.00	0.423 $\pm$ 0.03	0.233 $\pm$ 0.02	0.207 $\pm$ 0.02	0.491 $\pm$ 0.04	0.310	1.747	0.920
	MIST (Ours)	0.581$\pm$0.01	0.252$\pm$0.01	0.589$\pm$0.01	0.437$\pm$0.03	0.410$\pm$0.01	0.707$\pm$0.01	0.496	1.233	0.794
PCC@100	ST-Net	0.611 $\pm$ 0.01	0.221 $\pm$ 0.01	0.595 $\pm$ 0.00	0.355 $\pm$ 0.04	0.327 $\pm$ 0.01	0.696 $\pm$ 0.01	0.468	3.320	1.354
	Hist2ST	0.473 $\pm$ 0.07	0.274 $\pm$ 0.03	0.434 $\pm$ 0.03	0.222 $\pm$ 0.04	0.341 $\pm$ 0.03	0.446 $\pm$ 0.12	0.365	1.843	1.011
	BLEEP	0.632 $\pm$ 0.01	0.247 $\pm$ 0.02	0.659 $\pm$ 0.01	0.399 $\pm$ 0.04	0.377 $\pm$ 0.02	0.696 $\pm$ 0.01	0.502	1.558	0.915
	STEM	0.565 $\pm$ 0.01	0.192 $\pm$ 0.00 †	0.498 $\pm$ 0.03	0.236 $\pm$ 0.02	0.240 $\pm$ 0.03	0.602 $\pm$ 0.02	0.389	1.899	1.001
	MIST (Ours)	0.682$\pm$0.01	0.328$\pm$0.02	0.688$\pm$0.00	0.435$\pm$0.03	0.450$\pm$0.00	0.771$\pm$0.01	0.559	1.458	0.894

In the pooled setting, a single model is trained jointly across organs with slide-level hold-out and no organ labels. MIST again achieves the highest mean PCC on the six-organ set (0.599), ahead of BLEEP (0.587) and Hist2ST (0.556) (Table 3). Thus, for organs represented during training, MIST performs best both as a cohort-specific model and as one shared model.

Table 3: Cross-organ POOLED (six-organ subset). 5-fold slide-level holdout; mean_{std} over three seeds; best in bold.

Method	PCC $\uparrow$
ST-Net He et al. (2020) (NBE’20)	0.351 _{0.02}
Hist2ST Zeng et al. (2022) (BIB’22)	0.556 _{0.01}
BLEEP Xie et al. (2023) (NeurIPS’23)	0.587 _{0.00}
STEM Zhu et al. (2025) (ICLR’25)	0.273 _{0.00}
MIST	0.599_{0.01}

5.2 RQ2: Cross-organ transfer to an unseen organ

Having established strong performance on organs represented during training, we next evaluate transfer to an entirely unseen organ using leave-one-organ-out (LOOO) evaluation. Each fold withholds one organ from training and tests on that organ, ensuring that its anatomy is absent from the training set.

As shown in Table 4, MIST achieves the highest PCC on all six held-out organs and obtains the best mean PCC overall (0.422), ahead of BLEEP (0.398). This improvement is statistically significant (Wilcoxon $p=0.018$; Appendix E). Although performance decreases relative to the pooled setting (0.599), the consistent per-organ gains indicate that MIST transfers more effectively to unseen anatomy. A complementary mouse experiment in Appendix J shows the same trend.

Table 4: Cross-organ LOOO transfer (six-organ subset). Dataset columns report PCC as mean $\pm$ standard deviation over three seeds; each fold holds out one organ (leakage-safe panels, frozen UNI+CONCH). The final three columns average PCC, MSE, and MAE equally across the six held-out organs. Best per column in bold; $\uparrow$ higher is better and $\downarrow$ lower is better.

Model	IDC	CCRCC	LUNG	PAAD	PRAD	SKCM	PCC Avg. $\uparrow$	MSE Avg. $\downarrow$	MAE Avg. $\downarrow$
ST-Net	0.332 $\pm$ 0.02	0.074 $\pm$ 0.00	0.535 $\pm$ 0.07	0.382 $\pm$ 0.00	0.078 $\pm$ 0.01	0.500 $\pm$ 0.09	0.317	1.912	0.984
Hist2ST	0.350 $\pm$ 0.00	0.064 $\pm$ 0.01	0.581 $\pm$ 0.00	0.214 $\pm$ 0.17	0.062 $\pm$ 0.01	0.694 $\pm$ 0.02	0.328	1.356	0.876
BLEEP	0.401 $\pm$ 0.04	0.058 $\pm$ 0.00	0.630 $\pm$ 0.00	0.497 $\pm$ 0.03	0.091 $\pm$ 0.00	0.708 $\pm$ 0.01	0.398	1.410	0.847
STEM [†]	—	—	—	—	—	—	—	—	—
MIST (Ours)	0.425$\pm$0.03	0.098$\pm$0.01	0.687$\pm$0.02	0.499$\pm$0.03	0.105$\pm$0.01	0.720$\pm$0.03	0.422	1.288	0.849

[†]STEM values are being recomputed with the correct $T=1000$ DDPM schedule (an earlier run used $T=50$); results pending.

5.3 RQ3: Which components drive predictive accuracy?

Given MIST’s accuracy, we next ask which components drive it. To isolate each context stream, we run a factorial study over all eight combinations of local (L), global (G), and slide-level (S) context on the six-organ pooled set (Table 5). The full model (L + G + S) is best (PCC 0.605), and local attention is the critical stream: adding it (global + local) reaches 0.596, whereas dropping local in favor of global + slide (G + S) falls to 0.570. Crucially, replacing the spatial k NN neighbors with random ones lowers the full model from 0.605 to 0.577, showing that the benefit comes from spatial locality rather than added attention alone. This identifies local spatial-neighborhood aggregation as the primary contributor. Encoder-feature ablations are reported separately in Appendix G.

Table 5: Context-stream ablation (six-organ POOLED). Local (L), global (G), slide (S) streams; the “Full, random nbr.” row uses random rather than spatial- k NN neighbours. Mean_{std} over three seeds; best per column in bold.

Configuration	Streams	PCC $\uparrow$	MSE $\downarrow$	MAE $\downarrow$
Global + Local	L + G	0.596 $\pm$ 0.007	0.461 $\pm$ 0.024	0.414 $\pm$ 0.010
Global + Slide	G + S	0.570 $\pm$ 0.037	0.517 $\pm$ 0.130	0.439 $\pm$ 0.050
Full, random nbr.	L _{rand} + G + S	0.577 $\pm$ 0.009	0.466 $\pm$ 0.015	0.415 $\pm$ 0.004
MIST (ours)	L + G + S	0.605$\pm$0.006	0.421$\pm$0.011	0.400$\pm$0.013

5.4 RQ4: Does coordinate invariance hold in practice?

Table 6: Real-slide coordinate-transform check. The Table 1 CCRCC and PRAD UNI + CONCH HVG models (six and two held-out folds) and the Table 3 six-organ pooled UNI + CONCH model (five held-out folds; shared 14-gene panel) are evaluated across three trained seeds per setting, with image features fixed. Coordinates are translated by (37.2, -91.4), rotated by 0.731 radians, reflected across the vertical axis, or randomly permuted. PCC is averaged across folds, then reported as mean $\pm$ standard deviation across seeds. Each Δ PCC is relative to the original coordinates within the same setting; scores across gene panels are not directly comparable.

Coordinates	CCRCC (HVG; six folds)		PRAD (HVG; two folds)		Pooled (14 genes; five folds)	
	PCC $\uparrow$	Δ PCC	PCC $\uparrow$	Δ PCC	PCC $\uparrow$	Δ PCC
Original	0.25158 $\pm$ 0.01041	—	0.41020 $\pm$ 0.00730	—	0.59875 $\pm$ 0.00671	—
Translation	0.25157 $\pm$ 0.01027	-0.000014	0.41025 $\pm$ 0.00732	+0.000058	0.59871 $\pm$ 0.00668	-0.000036
Rotation	0.25175 $\pm$ 0.01053	+0.000166	0.41026 $\pm$ 0.00729	+0.000066	0.59876 $\pm$ 0.00670	+0.000010
Reflection	0.25158 $\pm$ 0.01041	0	0.41020 $\pm$ 0.00730	0	0.59875 $\pm$ 0.00671	0
Spot permutation	0.25154 $\pm$ 0.01029	-0.000035	0.41022 $\pm$ 0.00730	+0.000026	0.59870 $\pm$ 0.00674	-0.000049

We test coordinate transformations on held-out CCRCC and PRAD slides from the HVG experiments in Table 1, and on the six-organ pooled slides from Table 3. The original scores reproduce the reported results: 0.252 $\pm$ 0.01 for CCRCC, 0.410 $\pm$ 0.01 for PRAD, and 0.599 for the pooled setting. Across six CCRCC, two PRAD, and five pooled folds, with three trained seeds per setting, the three-seed mean PCC changes by at most 0.000166 after translation, rotation, reflection, or spot permutation (Table 6); the corresponding mean MSE and MAE change by at most 0.000128 and 0.000043. The pooled model uses its own shared 14-gene panel, not the intra-cohort HVG panel. Reflection leaves predictions exactly unchanged in this test. Translation, rotation, and permutation can change the k NN selection when distances tie or are numerically close: although aggregate accuracy is stable, individual predictions are not exactly invariant on these real slides.

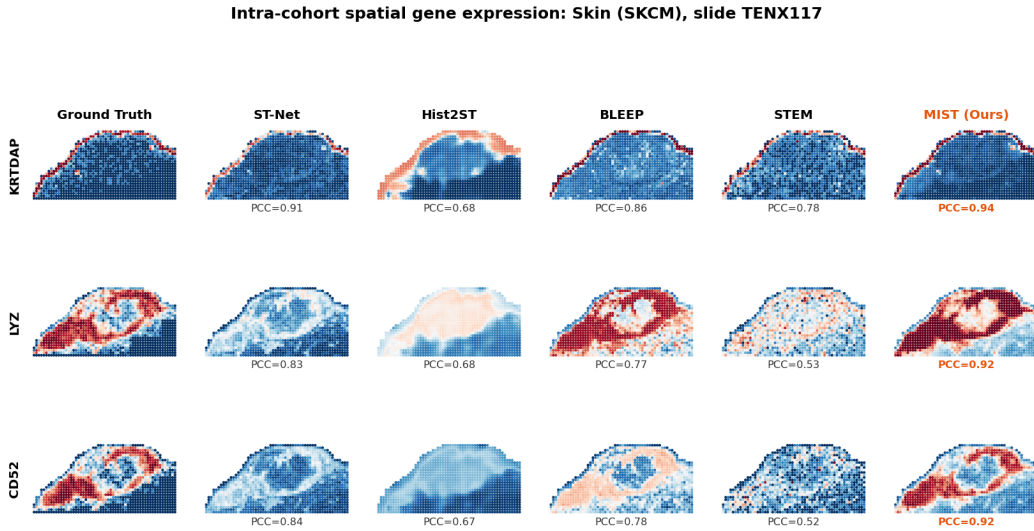


Figure 2: Qualitative spatial expression maps for a held-out skin (SKCM) slide (TENX117), three genes (rows). Each panel colors every spot by expression; MIST (right) most faithfully recovers the ground-truth spatial structure: the perivascular ring for LYZ/CD52 and the epidermal edge for KRTDAP. It attains the highest per-gene PCC (in orange), whereas STEM is noisy and Hist2ST over-smooths. Predictions use identical frozen UNI+CONCH features.

5.5 RQ5: Spatial structure preservation

Accurate per-gene prediction does not by itself guarantee that predictions preserve coherent tissue regions, so we finally ask whether MIST recovers spatial structure. To test this, we cluster spots (KMeans, $k=6$ domains) from either predicted or measured expression and compare the induced partitions using the adjusted Rand index (ARI) and normalized mutual information (NMI) on held-out slides of the six-organ subset (Table 10). MIST leads on both ARI (0.173) and NMI (0.242), ahead of the strongest baseline on each metric (BLEEP at 0.147 ARI, ST-Net at 0.201 NMI). Absolute scores vary widely across slides ($\text{std} \approx \text{mean}$), so the cohort means alone are hard to interpret; a paired per-slide comparison, however, is decisive (Figure 3): MIST beats BLEEP on 78% of slides for ARI and 90% for NMI ($n=58$ held-out slides; Wilcoxon $p < 0.001$ for both). MIST therefore recovers spatial domains more faithfully than the strongest baseline, even though the absolute agreement remains modest.

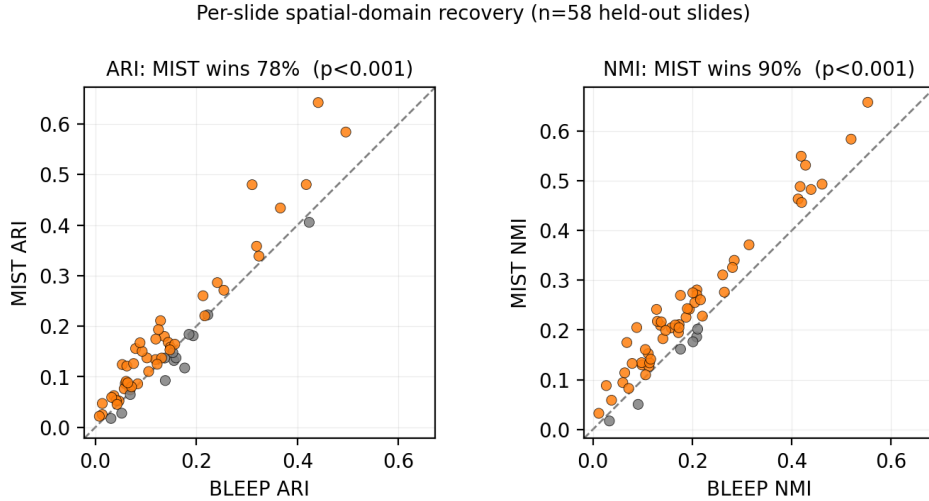


Figure 3: Per-slide spatial-domain recovery, MIST (y) vs. BLEEP (x) on the $n=58$ held-out slides. Each point is one slide; orange marks slides where MIST wins; the dashed line is $y=x$. Points spread widely (large across-slide variance) yet lie mostly above the diagonal, so MIST improves ARI on 78% and NMI on 90% of slides (Wilcoxon $p < 0.001$ for both).

5.6 Case Study: Recovering Cancer-Marker Genes

Average accuracy is clinically meaningful only if a model recovers the specific genes that define a cancer. We therefore test whether MIST predicts canonical marker genes for two cancers: melanoma (SKCM), through the melanocyte / MITF-pathway markers (MLANA, MITF, TYR, PMEL, DCT, TYRP1, S100B), and breast carcinoma (IDC), through the clinical immunohistochemistry panel (HER2/ERBB2, ER/ESR1, PR/PGR, GATA3, EPCAM, KRT8). Table 7 reports per-gene Pearson correlation on held-out slides. MIST recovers the melanoma markers almost perfectly (mean PCC 0.896; MLANA 0.948, MITF 0.943, TYR 0.936) and the breast markers strongly (mean 0.716; HER2 0.829), and it beats BLEEP on every marker in both cancers. Consistent with this, Figure 4 shows the predicted spatial maps reproduce the measured tumor architecture ($r \geq 0.91$ on the displayed slides). MIST thus reads out clinically actionable molecular markers, and their spatial localization, from H&E alone.

Table 7: Case study: per-gene recovery of canonical cancer markers. Per-gene Pearson $\uparrow$, mean over folds and three seeds; best per row in bold.

Cancer	Marker	MIST	BLEEP
Melanoma	MLANA	0.948	0.860
	MITF	0.943	0.862
	TYR	0.936	0.841
	S100B	0.888	0.823
	TYRP1	0.868	0.782
	PMEL	0.851	0.768
	DCT	0.842	0.790
	Mean	0.896	0.818
Breast	KRT8	0.881	0.846
	EPCAM	0.871	0.830
	HER2 (ERBB2)	0.829	0.750
	GATA3	0.829	0.786
	KRT5	0.625	0.567
	ER (ESR1)	0.580	0.505
	PR (PGR)	0.395	0.364
	Mean	0.716	0.664

Measured vs. MIST-predicted marker expression (held-out slides; dark = low, bright = high)

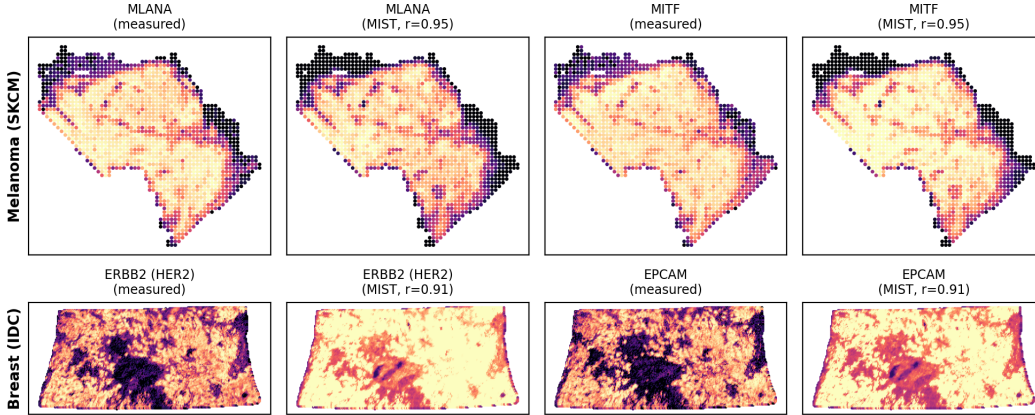


Figure 4: Spatial recovery of cancer markers. Measured vs. MIST-predicted expression on held-out melanoma (top) and breast (bottom) slides for representative markers (dark = low, bright = high). MIST reproduces the spatial architecture of each marker (r shown per panel).

Three findings merit emphasis. First, a compact direct predictor is competitive when features, gene panels, splits, and model selection are controlled; the 11.6% relative gain over the strongest baseline rests on matched experimental conditions, not a larger model. Second, pooled multi-organ performance is not a substitute for unseen-organ evaluation: the drop from 0.599 (pooled) to 0.422 (LOOO) shows that anatomical shift remains a separate and harder problem. Third, local spatial-neighborhood attention is the dominant contributor in the factorial study; global and slide pathways provide moderate complementary gains, and the complete model (L + G + S) achieves the best mean on the pooled evaluation.

Direct and generative models serve different purposes. STEM [Zhu et al. \(2025\)](#) models a conditional expression distribution and can represent one-to-many uncertainty through iterative sampling. MIST produces a point estimate optimized for correlation and squared

error. Its benefits, including deterministic, single-pass inference and explicit coordinate invariance, do not imply that morphology uniquely determines expression.

Limitations. The evaluation predicts 50 genes from frozen UNI+CONCH embeddings; results may not extend to transcriptome-scale output or end-to-end image encoding. LOOO uses a fixed benchmark panel and does not test platform or institution shift simultaneously. Coordinate invariance holds only for the spatial pathway with fixed image features; rotating the histology image can change its frozen embedding. Domain-recovery analysis covers only two baselines and has high slide-level variance. Pooled folds are sample-disjoint but cannot be verified as patient-disjoint where patient identifiers are unavailable.

6 Conclusion and Future Work

We developed MIST, a coordinate-invariant local-global spatial transformer for predicting gene expression from histology. Pairwise-distance encoding makes the spatial pathway invariant to rigid coordinate transformations by construction, while three parallel context signals: local k NN attention, global self-attention, and slide-level pooling, together produce complete-slide predictions in a single pass. These design choices directly address our design goals that multi-scale context is necessary and that architectural positional bias should not break under coordinate shift. Under matched HEST-1k evaluation, MIST improves mean intra-cohort correlation and remains competitive in LOOO transfer; the pooled-to-LOOO gap exposes the remaining difficulty of anatomical transfer as the central open challenge.

For future work, it would be valuable to explore how conditioning on slide-level metadata (organ type, sequencing platform) can narrow this anatomical gap, for instance, through FiLM or adaptive layer normalisation conditioners that are available at inference time without requiring expression labels. Scaling the gene panel beyond 50 genes, incorporating end-to-end image encoding, and combining MIST’s geometric structure with calibrated generative uncertainty modeling (as in STEM) are further natural directions.

AI use statement

Generative AI tools assisted with manuscript organization, language editing, ICLR formatting, and consistency checks between narrative claims and reported tables. They were not used to generate experimental data or execute model experiments. The authors reviewed all AI-assisted revisions and take responsibility for the final content of this work.

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Supplementary Material

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Table 8: Architectural components across representative histology to spatial transcriptomic (H&E to ST) predictors. • present, ○ absent.

Method	Local context	Global context	Slide token	Coord. invariant	Cont. distance
ST-Net He et al. (2020)	○	○	○	○	○
HisToGene Pang et al. (2021)	○	•	○	○	○
Hist2ST Zeng et al. (2022)	•	•	○	○	○
BLEEP Xie et al. (2023)	○	○	○	○	○
TRIPLEX Chung et al. (2024)	•	•	○	○	○
M2OST Wang et al. (2025)	•	○	○	○	○
STEM Zhu et al. (2025)	○	•	○	○	○
MIST (ours)	•	•	•	•	•

A Architectural Comparison

Table 8 summarizes architectural components across representative histology-to-spatial-transcriptomics predictors.

B Formal Guarantees

This section gives the detailed statements and proofs of the coordinate-frame invariance and permutation-equivariance properties used in the main text. The guarantees concern deterministic evaluation with fixed model parameters and fixed histology features. In finite-precision computation, negligible round-off differences may occur; nearest-neighbor distance ties must also be resolved consistently.

B.1 Coordinate-frame invariance

Proposition 1 (Coordinate frame invariance). Let $R \in \mathbb{R}^{2 \times 2}$ be orthogonal and let $\mathbf{t} \in \mathbb{R}^2$. For $\mathbf{P}' = \mathbf{P}R^\top + \mathbf{t}\mathbf{t}^\top$, the pairwise distances satisfy $D'_{ij} = D_{ij}$. Consequently, for fixed histology features $\mathbf{F}$,

$$f_\theta(\mathbf{F}, \mathbf{P}') = f_\theta(\mathbf{F}, \mathbf{P}). \quad (9)$$

Proof. For any two spots,

$$\mathbf{p}'_i - \mathbf{p}'_j = (\mathbf{p}_i - \mathbf{p}_j)R^\top, \quad (10)$$

$$\|\mathbf{p}'_i - \mathbf{p}'_j\|_2^2 = (\mathbf{p}_i - \mathbf{p}_j)R^\top R(\mathbf{p}_i - \mathbf{p}_j)^\top = \|\mathbf{p}_i - \mathbf{p}_j\|_2^2. \quad (11)$$

Thus, the full distance matrix and the distance-based k NN ordering are unchanged, assuming consistent tie handling. Any optional median-distance normalization is also unchanged because its input distances are unchanged, and therefore the RBF features $\phi(D_{ij})$ are unchanged. The local attention receives the same neighbor indices and distance features, while the global attention, slide pooling, residual blocks, and prediction head receive the same histology features and intermediate representations. Induction over the transformer layers gives identical outputs. The result does not assert invariance of the frozen image encoder to transformations of the H&E image itself. $\square$

B.2 Permutation equivariance

Proposition 2 (Permutation equivariance). Let $Q \in \{0, 1\}^{N \times N}$ be a permutation matrix. If the k NN construction resolves distance ties equivariantly, then

$$f_\theta(Q\mathbf{F}, Q\mathbf{P}) = Qf_\theta(\mathbf{F}, \mathbf{P}). \quad (12)$$

Proof. Permuting the rows of $\mathbf{P}$ permutes the rows and columns of its pairwise distance matrix. With equivariant tie handling, the neighbor list and corresponding distances are permuted in the same way. Hence each local attention computation is permuted with its

associated spot. The global self-attention operation is permutation-equivariant because queries, keys, and values are permuted together. Attention pooling produces the same pooled vector, since it is a weighted sum over the spots, while broadcasting that vector preserves the row permutation. The input projection, residual addition, layer normalization, feed-forward network, and final per-spot prediction head are all row-wise equivariant. Applying these observations inductively through all layers yields $f_\theta(Q\mathbf{F}, Q\mathbf{P}) = Qf_\theta(\mathbf{F}, \mathbf{P})$. $\square$

C Dataset

Table 9 lists the number of samples, total spots, and cross-validation folds for each HEST-1k cohort. The reported experiments use six HEST-1k cohorts: CCRCC (kidney, 24 samples / 74,220 spots), IDC (breast, 4 / 44,974), LUNG (lung, 2 / 7,505), PAAD (pancreas, 3 / 9,793), PRAD (prostate, 23 / 62,710), and SKCM (skin, 2 / 5,716).

Table 9: HEST-1k cohort statistics. Cohorts used in the reported experiments; spots are counted after barcode filtering and folds denote per-cohort train/test splits.

Cohort	Organ	Samples	Spots	Folds
IDC	Breast	4	44,974	4
CCRCC	Kidney	24	74,220	6
LUNG	Lung	2	7,505	2
PAAD	Pancreas	3	9,793	3
PRAD	Prostate	23	62,710	2
SKCM	Skin	2	5,716	2

Split granularity and patient disjointness. Each “sample” in Table 9 is a distinct HEST-1k slide (a whole-slide section), and every fold is slide-disjoint: no slide is shared between training and test, the pooled regime holds out entire slides, and the leave-one-organ-out regime holds out entire organs. HEST-1k patient/donor identifiers are not distributed with the per-slide data in our pipeline, so for a cohort in which one donor may contribute several slides we cannot verify patient-level disjointness. The intra-cohort and pooled numbers should therefore be read as slide-level generalization and could be optimistic if a cohort mixes multiple slides from the same patient; this risk is largest for the two multi-slide cohorts, CCRCC (24 slides) and PRAD (23 slides), and small for the remaining cohorts (IDC 4, PAAD 3, LUNG 2, SKCM 2 slides). Importantly, the cross-organ LOOO transfer results (our central generalization claim) are unaffected, because an entire organ (and hence a disjoint set of patients) is withheld at test time.

Table 10: Spatial-domain recovery (six-organ subset). ARI/NMI between domains (KMeans, $k=6$) from predicted vs. measured expression; mean_{std} over three seeds; best per column in bold.

Method	ARI $\uparrow$	NMI $\uparrow$
ST-Net He et al. (2020)	0.140 _{0.11}	0.201 _{0.12}
Hist2ST Zeng et al. (2022)	0.133 _{0.11}	0.180 _{0.11}
BLEEP Xie et al. (2023)	0.147 _{0.12}	0.197 _{0.13}
STEM Zhu et al. (2025)	0.101 _{0.12}	0.136 _{0.14}
MIST	0.173_{0.14}	0.242_{0.14}

D Extended Related Work

Machine-learning prediction of gene expression from histology. ST-Net [He et al. \(2020\)](#) established patch-level regression. Hist2ST [Zeng et al. \(2022\)](#) adds spatial context through a graph over neighboring spots, while BLEEP [Xie et al. \(2023\)](#) aligns image and expression

embeddings for cross-modal retrieval. STEM [Zhu et al. \(2025\)](#) instead treats expression prediction as conditional generation. These methods use different features, gene panels, splits, and selection procedures, so their published scores do not isolate architectural gains; our comparisons therefore use a matched protocol. MIST is a deterministic point predictor and does not model the full conditional expression distribution.

Geometric structure and coordinate invariance. Spatial prediction depends on the coordinate representation supplied to a model. Absolute position embeddings can encode location within a particular coordinate frame, but their output may change when an equivalent tissue layout is translated, rotated, or reflected. Geometric networks address this by constructing interactions from relative quantities [Satorras et al. \(2021\)](#). MIST uses pairwise Euclidean distances, which are preserved by translations, rotations, and reflections; the guarantee applies to the spatial pathway for fixed histology features, not to rotations of the image-encoder input.

Generalization across samples and organs. HEST-1k [Jaume et al. \(2024\)](#) spans many cohorts and organs. Within-cohort evaluation can exploit organ-specific morphology or acquisition effects. We therefore distinguish pooled cross-validation, which holds out samples while retaining their organ types in training, from LOOO, which withholds an entire organ and provides a stricter test of anatomical transfer.

Positioning of MIST. MIST connects these three directions: an inductive-bias-explicit spatial transformer for direct regression, evaluated under a matched cross-cohort protocol. Unlike absolute position encodings or frame-averaging strategies, its coordinate pathway is invariant by construction. Unlike generative models, it does not represent expression uncertainty; its advantage is deterministic, single-pass inference with explicit geometric structure. Unlike prior evaluations that report within-cohort accuracy only, our controlled comparison separates interpolation from anatomical transfer, making the difficulty of organ shift directly measurable.

E Paired Statistical Tests

This section collects the paired significance results reported in the main text. Tests compare MIST with the strongest relevant baseline under the same splits and evaluation protocol. The table reports the available aggregate statistics; complete per-run or per-slide effect-size vectors are not included in the manuscript.

Table 11: Summary of paired statistical tests. All tests are two-sided Wilcoxon signed-rank tests unless stated otherwise; p -values are reported as available from the matched evaluation.

Outcome	Comparison	Sample size	Reported result
PCC@50	MIST vs. BLEEP	$n = 18$ paired runs	gain 0.051; 95% bootstrap CI $[0.041, 0.062]$; $p = 7.6 \times 10^{-6}$
Per-gene PCC@50	MIST vs. BLEEP	$n = 300$ genes	MIST wins 87%; $p < 10^{-30}$
LOOO PCC	MIST vs. BLEEP	six held-out organs	$p = 0.018$
Spatial ARI	MIST vs. BLEEP	$n = 58$ slides	MIST wins 78%; $p < 0.001$
Spatial NMI	MIST vs. BLEEP	$n = 58$ slides	MIST wins 90%; $p < 0.001$

F Extended Experimental Setup

Dataset and preprocessing. We evaluate on HEST-1k [Jaume et al. \(2024\)](#) using the six cohorts listed in Table 9: CCRCC, IDC, LUNG, PAAD, PRAD, and SKCM. We use the

provided patient-stratified folds, $\log(1+x)$ normalization, and frozen UNI+CONCH features (1536-d). Unless stated otherwise, targets are the 50-gene HEST HVG panel. The panel study additionally uses matched 50-gene DEG (Wilcoxon selection across Leiden clusters) and HMHVG (mean-and-variability selection after variance filtering) panels (Table 1).

Evaluation protocol. Intra-cohort training fits a model within each cohort. Pooled five-fold validation trains across organs while holding out slide-level samples; organ identity defines the protocol but is not a model input. LOOO trains on all-but-one organ and tests the held-out organ. We report PCC, SCC, and MSE averaged over outer folds and three seeds. Checkpoints are selected on inner validation and evaluated once on the outer test fold; single-slide folds use disjoint spots for inner validation. Patient-level disjointness is not guaranteed where identifiers are unavailable.

Baselines and implementation. We compare with ST-Net [He et al. \(2020\)](#), Hist2ST [Zeng et al. \(2022\)](#), BLEEP [Xie et al. \(2023\)](#), and STEM [Zhu et al. \(2025\)](#) under matched features, panels, splits, budgets, and checkpoint selection. MIST uses four layers, width 256, four heads, eight neighbors, 16 radial bases, and 4.88M parameters. AdamW training uses learning rate 10^{-3} , weight decay 10^{-2} , cosine decay, gradient clipping, 100 epochs maximum, patience 20, and at most 3,000 spots per step.

G Encoder Ablation

We repeat the intra-cohort PCC@50 comparison (the 50-gene block of Table 2, which uses the concatenated UNI + CONCH features) with each foundation encoder in isolation: UNI only (ViT-L/16, 1024-d) and CONCH only (512-d), holding models, cohorts, gene panels, splits, and training budget fixed (Table 12). MIST remains the most accurate encoder-averaged method under both UNI-only and CONCH-only features.

Table 12: Encoder ablation: intra-cohort PCC@50 per encoder. UNI-only (1024-d) vs. CONCH-only (512-d); mean_{std} over three seeds; best per row in bold.

Encoder	Cohort	ST-Net	Hist2ST	BLEEP	STEM	MIST (Ours)
UNI (1024-d)	IDC Breast	0.538±0.008	0.250±0.067	0.508±0.005	0.436±0.005	0.569±0.009
	CCRCC Kidney	0.172±0.010	0.214±0.015	0.184±0.007	0.116±0.002	0.264±0.010
	LUNG Lung	0.482±0.012	0.378±0.097	0.559±0.007	0.393±0.009	0.590±0.005
	PAAD Pancreas	0.353±0.022	0.243±0.061	0.393±0.035	0.227±0.017	0.392±0.040
	PRAD Prostate	0.251±0.008	0.322±0.020	0.343±0.009	0.190±0.010	0.394±0.003
	SKCM Skin	0.632±0.010	0.510±0.015	0.639±0.015	0.471±0.031	0.694±0.001
	Average	0.405	0.319	0.438	0.305	0.484
CONCH (512-d)	IDC Breast	0.515±0.007	0.264±0.018	0.496±0.011	0.396±0.011	0.551±0.004
	CCRCC Kidney	0.160±0.010	0.126±0.026	0.141±0.012	0.078±0.001	0.222±0.010
	LUNG Lung	0.517±0.001	0.435±0.006	0.576±0.003	0.386±0.030	0.554±0.005
	PAAD Pancreas	0.372±0.035	0.186±0.035	0.384±0.033	0.251±0.028	0.387±0.045
	PRAD Prostate	0.327±0.003	0.322±0.009	0.364±0.010	0.210±0.013	0.406±0.006
	SKCM Skin	0.633±0.002	0.388±0.022	0.633±0.005	0.453±0.014	0.658±0.007
	Average	0.421	0.287	0.432	0.296	0.463

H Complete LOOO (Leave-One-Organ-Out) Results

Table 4 reports the complete per-cohort leave-one-organ-out transfer results on the six-organ subset summarised in the main text (Section 5.2, Figure 5). Each fold holds out one entire organ and trains on the other five, using leakage-safe per-fold gene panels and identical frozen UNI + CONCH features. MIST wins every held-out cohort and the average (0.422).

Table 13: Cross-organ LOOO transfer (six-organ subset). Per-cohort Pearson $\uparrow$; each fold holds out one organ (leakage-safe panels, frozen UNI + CONCH); mean_{std} over three seeds; best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	BLEEP NeurIPS'23	STEM ICLR'25 [†]	MIST (Ours)
CCRCC Kidney	0.074 $\pm$ 0.00	0.064 $\pm$ 0.01	0.058 $\pm$ 0.00	0.024 $\pm$ 0.01	0.098$\pm$0.01
IDC Breast	0.332 $\pm$ 0.02	0.350 $\pm$ 0.00	0.401 $\pm$ 0.04	0.131 $\pm$ 0.08	0.425$\pm$0.03
LUNG Lung	0.535 $\pm$ 0.07	0.581 $\pm$ 0.00	0.630 $\pm$ 0.00	0.067 $\pm$ 0.00	0.687$\pm$0.02
PAAD Pancreas	0.382 $\pm$ 0.00	0.214 $\pm$ 0.17	0.497 $\pm$ 0.03	0.052 $\pm$ 0.00	0.499$\pm$0.03
PRAD Prostate	0.078 $\pm$ 0.01	0.062 $\pm$ 0.01	0.091 $\pm$ 0.00	0.016 $\pm$ 0.00	0.105$\pm$0.01
SKCM Skin	0.500 $\pm$ 0.09	0.694 $\pm$ 0.02	0.708 $\pm$ 0.01	0.299 $\pm$ 0.00	0.720$\pm$0.03
Average	0.317	0.328	0.398	0.098	0.422

[†]STEM incomplete (1–2 of 3 seeds); values provisional.

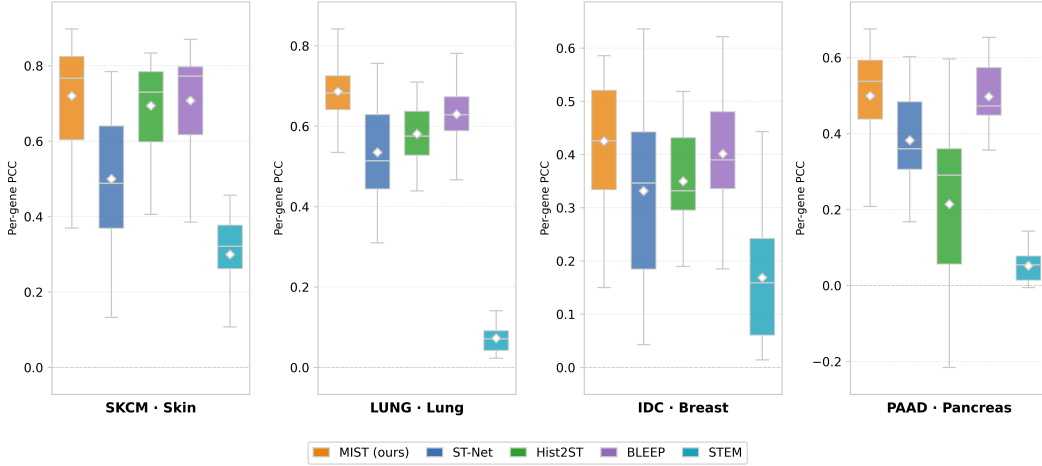


Figure 5: Per-cohort cross-organ LOOO accuracy (four representative held-out cohorts; kidney and prostate omitted for readability, frozen UNI+CONCH). One panel per held-out cohort; box = per-gene PCC over three seeds, $\diamond$ = mean. MIST (orange) leads in all four panels, and wins all six held-out organs in Table 4.

I Computation Resources, Model Complexity, and Inference Scaling

All experiments ran on a shared GPU cluster. Feature extraction and training used NVIDIA GPUs with compute capability $\geq \text{sm_80}$ (A100 80GB, A100-PCIe, L40s, and H200); Volta (V100) GPUs are incompatible with our CUDA-13 / PyTorch build and were excluded.

- Encoders: frozen UNI (ViT-L/16, 1024-d) and CONCH (512-d) patch features, concatenated to 1536-d; no encoder fine-tuning.
- Model: MIST has 4.88M trainable parameters (Table 14) and predicts a whole slide in a single forward pass, with no iterative sampling.
- Training: AdamW with a cosine schedule, 100 epochs with early stopping (patience 20), nested validation, and a correlation-aware objective; three seeds per configuration.
- Inference: single-pass; median model-only latency and peak memory versus slide size are reported in Table 15, with feature extraction and disk I/O excluded.

Table 14: Model complexity. Parameter counts and inference mechanisms.

Model	#Params (M)	Inference
ST-Net He et al. (2020) (NBE’20)	0.47	single-pass
Hist2ST Zeng et al. (2022) (BIB’22)	2.18	single-pass
BLEEP Xie et al. (2023) (NeurIPS’23)	0.54	single-pass
STEM Zhu et al. (2025) (ICLR’25)	34.16	iterative
MIST (V3)	4.88	single-pass

Table 15: MIST inference scaling (A100 80GB). Median model-only latency and peak GPU memory vs. slide size; feature extraction and I/O excluded.

Spots	Latency (ms)	Memory (MiB)
500	3.48	43
1,000	3.63	67
3,000	9.36	335
5,000	21.31	842
10,000	64.74	3,183

J Mouse cross-organ diagnostic

We evaluate 59 mouse Visium slides spanning seven organs with mouse-specific 50-gene panels. Pooled performance reaches 0.887, whereas every method remains near the floor under LOOO; we therefore do not interpret the latter as evidence of successful cross-species organ transfer.